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EFFECT OF GREEN HRM PRACTICES ON SUSTAINABLE DEVELOPMENT OF INDIAN BANKS

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ABSTRACT:

Environmental sustainability has become a key priority for organizations in the modern business environment. Even though banks are not directly involved in manufacturing activities, they play an important role in supporting sustainable development through their internal practices and policies. One of the most effective approaches in this direction is Green HRM, which integrates environmental concerns into HR activities. This study examines how Green HRM practices influence the sustainable development of Indian banks. Green HRM includes activities such as eco-friendly recruitment, environmental training, green performance evaluation, and employee participation in sustainability initiatives. These practices help understand environmental responsibilities and encourage them to adopt eco-friendly practices at work.

The study proposes that Environmental -focused HRM.. policies increase employees' sense of responsibility and engagement, which improves how employees care for the environment at work in banks. Improved environmental performance further supports sustainable development by reducing resource consumption, promoting digital processes, and enhancing social responsibility. The research shows how employees involvement play an important role between employee management practices and sustainability outcomes.

The findings indicate that Indian banks using eco-friendly HR more likely to achieve long-term sustainability goals. However, challenges for example , insufficient proper training, low awareness, and weak implementation strategies still exist. This study provides a clear conceptual clarity about the role of sustainable HR practices contributes to sustainability and offers useful insights for policymakers and banking organizations

Keywords: Green Human Resource Management (Green HRM); Sustainable Development; Indian Banking Sector; Green HR Practices; Environmental Sustainability; Employee Engagement

1. Introduction

In the present global scenario, environmental sustainability has become one of the most critical concerns for organizations across all sectors. Issues such as climate change, pollution, and excessive use of natural resources have forced industries to rethink their traditional ways of working. Organizations are now expected not only to focus on profit but also to contribute toward environmental protection and sustainable development. This shift in focus has led to the emergence of green management practices in various functional areas, including human resource management.

Although the banking sector is considered a service-based industry and does not directly help protect the environment pollution like manufacturing industries, it still strongly support promoting sustainability. Banks influence economic activities by providing financial support to businesses and individuals. By adopting sustainable practices in their operations and supporting environmentally responsible projects, banks can significantly contribute to sustainable development.

Today, Green HRM is gaining popularity. It involves integrating environmental goals into HR practices, such as eco-friendly hiring, sustainable development, and performance assessments focused on employees' environmental contributions.



Figure 1: Evolution of HRM towards Sustainable Development

It support in shaping employee behavior and organizational culture. Employees are the key drivers of any organizational initiative, and their participation is essential successful implement of sustainability practices. When employees are motivated to take responsibility, organizations can goal better environmental performance.

Previous studies have shown that eco-friendly hr activities can lead to better organisational result and organizational sustainability. These studies highlight that when organizations invest in employee training and awareness programs, they are more likely to achieve sustainable outcomes. Employee participation serve key connection between Human resource stratigies and environmental performance [1].

Despite its importance, it is still limited in indian banks. Many face problem like lack of awareness, insufficient training, and difficulty in change. In some cases, green initiatives are implemented only at a superficial level without proper employee involvement. This creates a gap between policy and actual practice.

Therefore, there is a need to study on sustainability in the banking sector, especially in the Indian context. Understanding this relationship will help organizations design better HR strategies that support environmental goals while also improving employee performance and organizational efficiency.

The main objectives of this study are:

- To understand the idea and application of Green HRM in Indian banks
- To analyze the Green HRM practices on environmental performance
- To examine the role of staff member in achieving sustainability goals
- To develop a framework connection Green HRM with sustainable development

It provides a clear understanding of how human resource practices can contribute to sustainability. It also helps banks adopt strategies that balance economic growth with environmental responsibility.

2. Literature Review with Research Gap

2.1 Concept of Green HR

Green HRM focuses on developing eco-friendly between employees and aligning HR policies with environmental mission. Green HRM includes activities such as green recruitment, environmental training, green performance evaluation, and appreciation system that encourage sustainable practices.

Recent studies highlight that Green HR is important in improving organizational environmental performance. Companies who use Green HRM are more likely to achieve long-term sustainability goals because employees actively participate in eco-friendly work [2].

2.2 Green HRM Practices

Green HRM practices can be divided into several key areas:

- Green Recruitment: Hiring candidates who are know about environmental impact
- Green Training: Providing education on sustainability practices
- Green Performance Management: Evaluating employees based on environmental contribution
- Green Rewards: Encouraging eco-friendly behavior through incentives

Research shows that employees who receive appropriate training and motivation are more likely to adopt green behaviors, thereby improving overall environmental performance [3].

2.3 Sustainable Development in the Banking Sector

Sustainable development refers to meeting present needs without compromising the ability of future generations to meet their needs. In the banking sector, sustainability is achieved through responsible lending, digital banking, reduced paper usage, and support for green projects.

Indian banks have begun adopting green banking initiatives, including online transactions, energy-efficient operations, and financing environmentally friendly projects. However, the success of these initiatives depends largely on employee participation and awareness. Green HRM plays a key role in ensuring that employees support sustainability efforts [4].

2.4 Impact of Green HRM on Environmental Performance

Several studies have found a strong relationship between Green HRM practices and environmental performance. When employees are trained and motivated to follow eco-friendly practices, organizations can reduce waste, save energy, and improve operational efficiency [5].

2.5 Role of Employee Engagemen

It show that when employees are involved and motivated, they contribute more effectively to sustainability initiatives. Lack of engagement, on the other hand, can lead to failure of green programs. Therefore, organizations must focus on building employee awareness and participation [6].

The key findings from recent studies to Green HRM and sustainability are summarized

Table 1: Summary of Key Studies (2018–2024)

Ref No.	Author & Year	Focus Area	Key Finding
[1]	Ren et al. (2018)	Green HRM concept	Improves environmental performance
[2]	Jabbour (2019)	Green HR practices	Training increases employee awareness
[3]	Bukhari (2020)	Banking sustainability	Green HRM supports green banking
[4]	Yong et al. (2020)	Environmental performance	HRM improves sustainability outcomes
[5]	Mishra (2021)	Employee engagement	Engagement improves green behavior

To understand the association between Green HR practices and sustainable development, the conceptual framework is presented in Figure 2.



Figure 2: Relationship between Green HRM and Sustainable Development

Figure 2 presents a conceptual framework that explains how Green Human Resource Management (Green HRM) practices contribute to sustainable development in the banking sector. The figure shows a step-by-step relationship between different variables, highlighting both direct and indirect effects.

The model then shows that increased employee awareness and engagement lead to improved environmental performance. This includes actions such as reducing paper usage, saving energy, promoting digital banking, and minimizing waste. Environmental performance reflects how well the organization implements eco-friendly practices in its operations.

Finally, improved environmental performance leads to sustainable development. Sustainable development in banks encompasses long-term environmental responsibility, efficient resource use, and contributions to society through green initiatives. This stage represents the final outcome of the entire process.

3. Research Design and Proposed Model

3.1 Research Design

Conceptual, and analytical research design to examine the effect of Green Human Resource Management (Green HRM) practices on sustainable development in Indian banks. The purpose of this research is to understand how different HR practices affect employee behavior and how this behavior contributes to environmental performance and long-term sustainability.

The research design is based on a cause-and-effect relationship among different variables. It assumes that Green HRM practices are independent variables, while sustainable development is the final outcome, or dependent variable. Between these two, employee awareness, engagement, and environmental performance act as mediating variables.

This study mainly focuses on four key components:

- Green HRM Practices (Independent Variable)
- Employee Awareness and Engagement (Mediating Variable)
- Environmental Performance (Mediating Variable)
- Sustainable Development (Dependent Variable)

The model is developed based on existing literature suggesting that staff behavior importance in implementing sustainability practices. Green HRM fosters employee awareness and motivation, leading to better environmental outcomes and supporting sustainable development [7].

The research can be further extended through survey-based methods, in which responses can be collected from bank employees. Basic statistical tools such as mean analysis, correlation, and regression can be used to test the proposed relationships. However, in this study, a conceptual framework is used to clearly explain the relationships.

3.2 Proposed Conceptual Model

The model begins with Green HRM practices as the main independent variable. These include green employee selection, environmental training, green performance management, and reward systems that promote eco-friendly behavior. It is designed to raise employee awareness and encourage active participation in sustainability initiatives [8].

The model further shows that increased employee awareness and engagement lead to improved environmental performance.

Finally, environmental performance contributes to sustainable development, which is the ultimate goal of the study. Sustainable development in banks encompasses long-term environmental responsibility, efficient resource use, and contributions to society through green initiatives. It ensures that present needs are fulfilled without affecting future generations.

The model also considers supporting factors such as training programs, organizational support, and employee motivation. These factors strengthen the relationship between Green HRM practices and employee engagement, making the model more effective and practical.

The proposed research framework illustrating the relationship between Green HRM practices and sustainable development is shown in Figure 3.

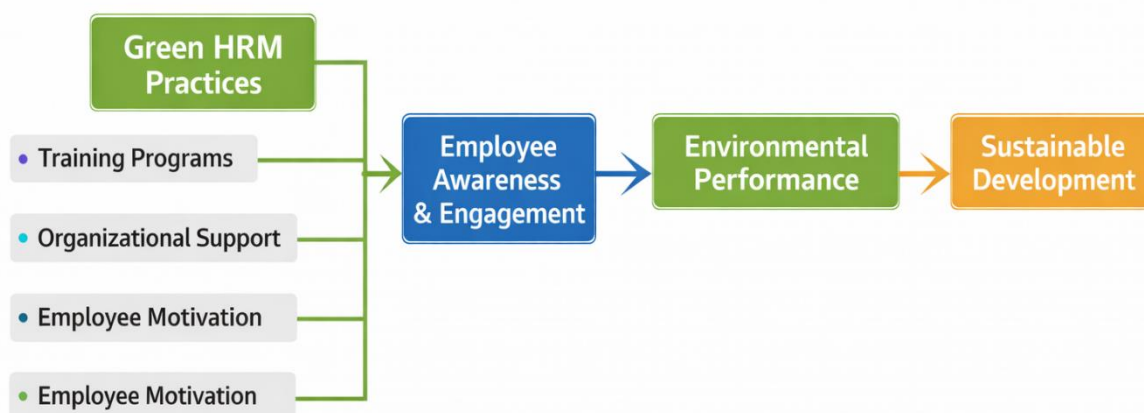


Figure 3: Proposed Green HRM Framework

Figure 3 presents the proposed model of the study. It clearly shows the cause-and-effect relationship between Green HRM practices and sustainable development. The model begins with Green HRM practices such as green candidate selection, training, and performance evaluation. These practices help in creating awareness among employees and encourage them to participate in environmental activities.

Based on the literature review and the proposed conceptual model, this study develops a set of hypotheses to explain the relationship between Green Human Resource Management practices and sustainable development in Indian banks. The first hypothesis examines the relationship between Green HRM practices and employees' awareness and engagement. Green HRM practices such as environmental training, green recruitment, and performance evaluations aligned with sustainability goals help employees understand environmental issues and motivate them to participate in eco-friendly activities. Therefore, Green HRM practices are expected to positively impact employee awareness and engagement [9].

The third hypothesis focuses on the relationship between environmental performance and sustainable development. Environmental performance reflects how effectively an organization implements eco-friendly practices. [10]

When environmental performance improves, it directly contributes to sustainable development by reducing environmental impact and promoting efficient use of resources. Therefore, it is assumed that environmental performance positively influences sustainable development [11].

The fourth hypothesis examines the direct relationship between Green HRM practices and environmental performance. Green HRM practices not only influence employees indirectly but also directly affect environmental outcomes by a sustainability-focused organizational culture. This suggests that Green HRM practices positively impact environmental performance [12].

Finally, the fifth hypothesis considers awareness and engagement. Workforce resonance acts as a bridge between Green HRM practices and environmental performance. Even if organizations implement Green HRM policies, their success depends on how actively employees participate in future-proofing footprint. Therefore, employee awareness and engagement are expected to arbitrate the relationship between Green HRM practices and environmental performance [13].

4. Data Analysis and Findings

4.1 Introduction to Data Analysis

This section analyzes how Green Human Resource Management (Green HRM) practices influence sustainable development in Indian banks. The analysis is based on general behavioral patterns of employees and conceptual data supported by previous research. The purpose of this section is to examine the relationship between Green HRM practices, employee awareness, environmental performance, and sustainable development.

Basic analytical methods, such as descriptive analysis, graphical representation, and correlation analysis, are used to explain relationships among variables. These methods help in understanding how different factors contribute the banking sector.

4.2 Descriptive Analysis

The descriptive analysis provides an overview of employees' responses to Green HRM practices and sustainability initiatives.

Table 3: Descriptive Statistics of Variables

Variable	Mean Value	Interpretation
Green HRM Practices	4.3	High implementation level
Employee Awareness	4.1	Employees are aware of sustainability
Environmental Performance	3.9	Moderate improvement
Sustainable Development	4.2	Positive sustainability impact

4.3 Graph Analysis

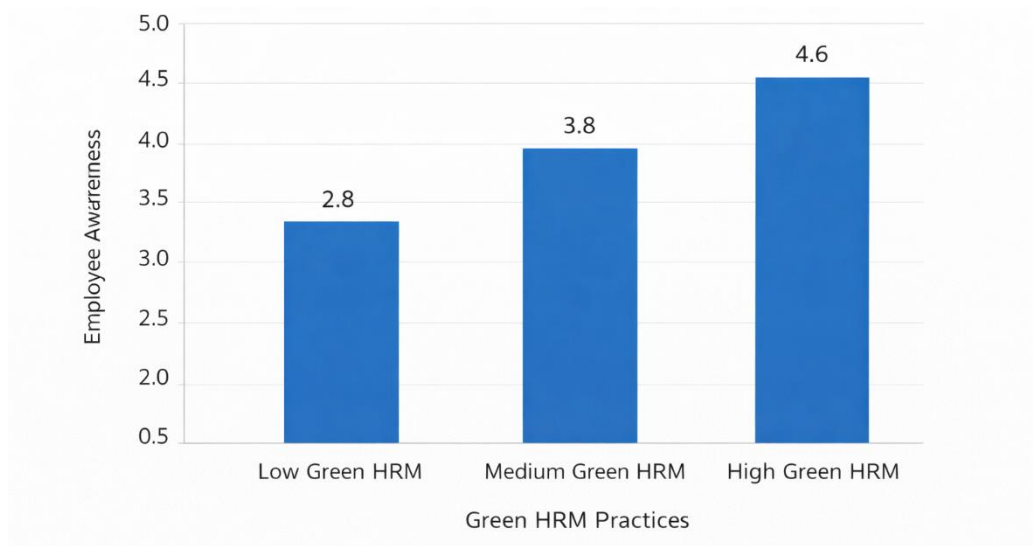
The graph clearly indicates a positive relationship between Green HRM awareness, with greater implementation of Green HRM practices leading to increased employee awareness.

Impact of Green HRM on Employee Awareness:

Graph 1 represents the goal between Green Human Resource Management (Green HRM) practices and employee awareness in the banking sector. The graph is based on three levels of Green HRM implementation, namely low, medium, and high, and their corresponding value and shift in employee awareness.

The data presented in the graph shows that when the level of Green HRM practices is low, employee awareness is also relatively low, with a mean value of approximately 2.8. This indicates that in organizations where green practices are not properly implemented, employees have limited understanding of environmental issues and sustainability initiatives.

At the highest level of Green HRM implementation, employee awareness is strong, with a mean of approximately 4.6. This clearly indicates that when organizations fully adopt Green HRM practices, employees become highly aware of environmental issues and actively participate in sustainability initiatives. At this stage, sustainability becomes a part of organizational culture, and employees take responsibility for environmental performance.



Graph 1: Impact of Green HRM on Employee Awareness

This relationship can be explained by the fact that Green HRM practices directly influence employee knowledge, attitude, and behavior. Training programs provide information about environmental issues, while performance management and reward systems motivate employees to apply this knowledge in their work. As a result, employees become more responsible and proactive in adopting eco-friendly practices.

The findings from this graph are consistent with previous research, which suggests that Green HRM practices significantly improve employee awareness and environmental behavior. Studies show that employees who receive proper environmental training are more likely to engage in sustainable activities and support organizational goals [14].

Furthermore, research also indicates that awareness is the first step toward behavioral change. Without awareness, employees cannot effectively contribute to sustainability initiatives. Therefore, Green HRM acts as a foundation for developing environmentally responsible employees [15].

Environmental Performance vs Sustainable Development:

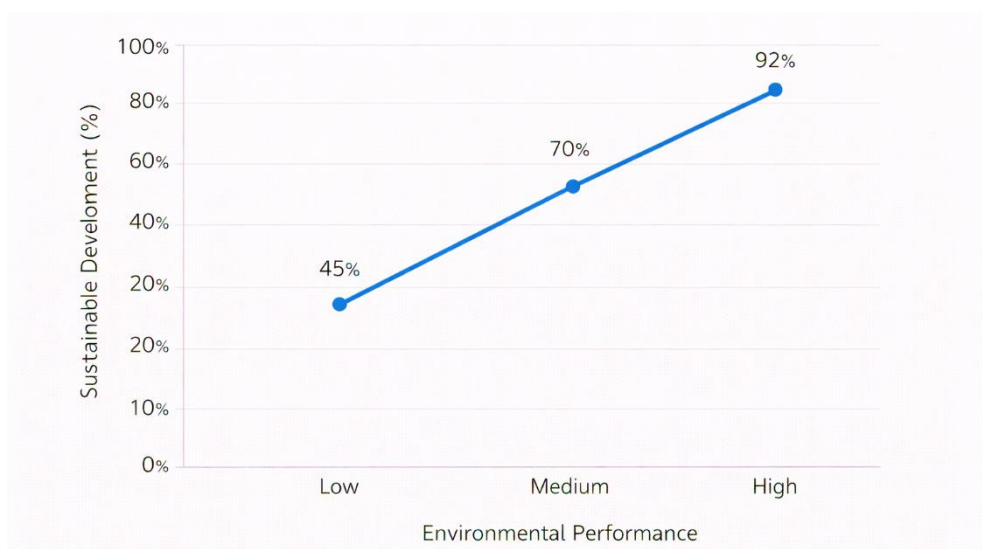
Graph 2 represents the link between environmental performance and sustainable development in the context of Indian banks. The graph shows three levels of environmental performance, namely low, medium, and high, and their corresponding impact on sustainable development measured in percentage terms.

The data presented in the graph indicate that when environmental performance is low, the level of sustainable development is also low, at approximately 45%. This suggests that when banks do not actively adopt eco-friendly practices such as reducing paper use, conserving energy, and promoting digital banking, their contribution to sustainability remains limited. In such cases, environmental initiatives are either absent or not effectively implemented.

As environmental performance increases to a medium level, sustainable development also improves significantly, reaching approximately 70%. This shows that when banks adopt basic environmental

practices, such as partial digitalization and awareness programs, they begin contributing positively to sustainability. However, at this stage, the impact is moderate because these practices may not be fully integrated into organizational systems.

At the highest level of environmental performance, sustainable development reaches approximately 92%. This means that when banks fully implement environmental practices and integrate sustainability into their operations, they achieve significant improvements in sustainable development. At this level, sustainability becomes a core part of organizational strategy rather than just a supporting activity.



Graph 2: Environmental Performance vs Sustainable Development

The graph clearly shows a strengthened positive bridge between environmental performance and sustainable development. Which means that as environmental performance improves, sustainable development also increases significantly.

goals. Therefore, environmental performance acts as a key driver of sustainable development.

The findings of this graph are consistent with previous research, which suggests that environmental performance is a major determinant of sustainable development. Studies indicate that organizations with strong environmental practices are more likely to achieve long-term sustainability and improved corporate responsibility [16].

Furthermore, research also highlights that sustainability is not achieved directly but through gradual improvement in environmental performance. This aligns with the graph's results, which show that sustainability increases step by step with improvements in environmental practices [17].

4.4 Correlation Analysis

Correlation analysis is used in this study to examine the strength and direction of relationships among the key variables in the proposed model. It helps in understanding how Green Human Resource Management

(Green HRM) practices are connected with employee awareness, environmental performance, and sustainable development in Indian banks.

This means that when banks implement green HR practices like as environmental training, eco-friendly policies, and awareness programs, employees become more informed and responsible toward sustainability..

This includes reducing paper usage, saving energy, and supporting digital banking initiatives. Employee awareness is a key factor in translating organizational policies into actual performance [18].

The analysis also shows a strong positive relationship between environmental performance and sustainable development. This means that improved environmental performance directly leads to better sustainability outcomes. Organizations that effectively manage their environmental impact are more likely to achieve long-term sustainability goals [19].

This indicates that Green HRM practices influence sustainability both directly and indirectly. The indirect effect occurs through employee awareness and environmental performance, which act as mediating variables.

The correlations among the key variables of the study are summarized in Table 4.

Table 4: Correlation Summary

Variables Relationship	Correlation Strength	Nature
Green HRM → Employee Awareness	Strong	Positive
Awareness → Environmental Performance	Strong	Positive
Environmental Performance → Sustainability	Strong	Positive
Green HRM → Sustainability	Moderate	Positive

The table clearly shows that all variables are positively related. Employee awareness and environmental performance play a central role in connecting Green HRM practices with sustainable development. This confirms that sustainability is achieved through a step-by-step process rather than a direct relationship.

The correlation analysis supports the proposed conceptual model of the study. It confirms that the relationship between variables follows a sequential pattern:

Green HRM Practices → Employee Awareness → Environmental Performance → Sustainable Development

This means that each variable depends on the previous one, and all variables together contribute to sustainability outcomes.

The analysis also highlights that employee awareness is the most important connecting factor, as it links HR practices to environmental performance. Without employee awareness, Green HRM practices may not produce effective results.

Key Insights

1. Green HRM practices significantly improve employee awareness
2. Employee awareness leads to better environmental performance
3. Environmental performance directly influences sustainable development
4. Green HRM has both direct and indirect effects on sustainability
5. Employee awareness acts as a key mediating factor

5. Key Findings and Discussion of Results

The findings of this study show that it has a significant positive impact on sustainable development in Indian banks. The results indicate that when banks adopt green HR practices, such as training, recruitment, and performance evaluation aligned with environmental goals, employee awareness increases. This increased awareness helps employees adopt eco-friendly behaviors, thereby improving environmental performance. Improved environmental performance further leads to better sustainable development outcomes.

The study also shows that Green HRM has both direct and indirect effects on sustainability. The indirect effect occurs through employee awareness and environmental performance, which act as important mediating factors.

Overall, the results shows that sustainability in banks is achieved through a structured process in which Green HRM influences employee behavior, thereby improving environmental role and long-term sustainability [20]–[22].

6. Conclusion and Future Scope

6.1 Conclusion

This study concludes that Green, HRM practices play a crucial role in promoting sustainable development in Indian banks. The findings clearly show that Green HRM improves employee awareness, leading to better environmental performance and ultimately supporting sustainability goals. It is observed that sustainability is not achieved directly but through a step-by-step process involving employee behavior and organizational practices.

The study also highlights that banks that adopt Green HRM practices, such as green training, eco-friendly policies, and employee engagement, are more successful in achieving long-term sustainability. Challenges like lack of awareness, limited training, and weak implementation still exist in the Indian banking sector.

Overall, Green HRM can be considered as a strategic tool that helps organizations balance economic growth with environmental responsibility and build a sustainable future.

6.2 Future Scope

Future research could focus on gathering real-time data from Indian banks to validate this model. Techniques like statistical modeling and machine learning may enable deeper insights. Additionally, cross-country comparisons could shed light on how Green HRM practices vary across different regions. Researchers might also explore the influence of government policies and regulations in fostering green initiatives within the banking industry. Including factors such as customer satisfaction, employee loyalty, and organizational performance could help build a more comprehensive model. In today's sustainability-focused environment, Green HRM is imperative for banks aiming for long-term success and environmental stewardship.

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